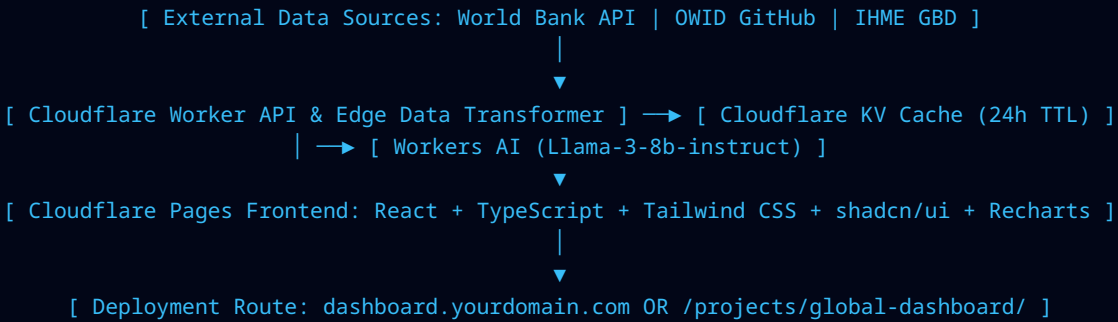


TECHNICAL SPECIFICATION & IMPLEMENTATION SCOPE (V2 REFINED)

Global Intelligence Dashboard

A modular, production-grade analytics platform blending macro economics, clean energy transition, and public health metrics powered by Cloudflare Workers AI.

1. System Architecture Blueprint



2. Workspace Setup & Directory Layout

Construct a clean monorepo/workspace structure to maintain full modular separation between your portfolio host and this standalone project.

```
global-dashboard/
├── apps/
│   ├── web/           # Vite + React + Tailwind + shadcn/ui + Recharts
│   └── worker/        # Cloudflare Worker Edge Transformer & AI Engine
├── data/
│   └── raw/           # Local dev cache for raw IHME CSV files
├── package.json       # Monorepo / workspace configuration
└── README.md
```

Recommended VS Code Tooling:

TAILWIND CSS INTELLISENSE

ESLINT

CLOUDFLARE WRANGLER

PRETTIER

3. Data Architecture & Source Blending Strategy

Data cleaning, join logic, and normalization happen inside the **Cloudflare Worker at the edge**. The frontend requests a single, pre-blended payload per entity/year mapped on **ISO 3166-1 alpha-3** and **Time (Year)**.

Data Source	Target Metrics Included	Ingestion Method	Format
World Bank Open Data	GDP per capita, Urban Population %, Internet Penetration %	Dynamic REST API fetch	JSON
Our World in Data (OWID)	Primary Energy Consumption, Renewable Energy Share %	GitHub Raw fetch / Static CSV	JSON / CSV
IHME (GBD Study)			CSV

Data Source	Target Metrics Included	Ingestion Method	Format
	Life Expectancy at Birth, DALYs (Disability-Adjusted Life Years)	Pre-processed export pipeline	

Unified Data Schema (TypeScript Interface)

```
export interface GlobalEntityMetric {
  countryCode: string; // e.g., "USA", "BRA", "KEN"
  countryName: string;
  year: number;
  economic: {
    gdpPerCapita: number | null;
    internetPenetration: number | null;
  };
  energy: {
    renewableShare: number | null;
    primaryEnergyPerCapita: number | null;
  };
  health: {
    lifeExpectancy: number | null;
    dalysRate: number | null;
  };
}
```

4. Backend API & Cost-Effective AI Layer

Using **Cloudflare Workers** combined with **Cloudflare Workers AI** provides a zero-overhead, highly performant edge layer that runs LLM inference for free within Cloudflare's daily allowance.

Cloudflare Worker AI Endpoint Implementation

```
// apps/worker/src/ai.ts
export async function generateMetricSummary(
  env: Env,
  countryName: string,
  dataPayload: object
): Promise<string> {
  const prompt = `
    You are an expert economic and public health analyst.
    Analyze the following merged dataset for ${countryName}:
    ${JSON.stringify(dataPayload)}

    Provide a concise, 3-bullet-point summary highlighting key trends between wealth (GDP), clean energy ad
  `;

  const response = await env.AI.run('@cf/meta/llama-3-8b-instruct', {
    messages: [
      { role: 'system', content: 'You are a concise data story analytics assistant.' },
      { role: 'user', content: prompt }
    ],
    max_tokens: 250
  });

  return response.response;
}
```

Caching Strategy: Store fetched external API results in **Cloudflare KV** with a 24-hour expiration (`expirationTtl: 86400`). This keeps dashboard load speeds under 150ms and prevents external rate limiting.

5. Frontend UI & Dashboard Component Architecture

The UI is constructed using **shadcn/ui primitives** and **Recharts** for a sustainable, fully component-owned frontend codebase.

Region A: Global Controls

- Country Search / Selector dropdown
- Multi-year timeline slider (1990–2026)
- Metric normalizer (Per Capita toggles)

Region C: Primary Visuals

- **Scatter Plot:** Wealth vs. Health correlation
- **Dual-Axis Line Chart:** Energy vs. DALYs over time
- **Choropleth Map:** Regional metric heatmaps

Region B: KPI Metric Cards

- GDP per Capita with YoY Delta %
- Renewable Energy Share % indicator
- Life Expectancy & DALYs Health score

Region D: AI Insights Panel

- On-demand Llama-3 summary generation
- Glowing pulse loading states
- Markdown summary renderer

6. Cloudflare Publishing & Modular Integration

1. Deploy Edge Backend API

```
cd apps/worker
npx wrangler deploy
```

2. Deploy Frontend Static Build

```
cd apps/web
npm run build
npx wrangler pages deploy dist --project-name=global-analytics-dashboard
```

3. Modular Routing Strategies

- **Option A (Dedicated Subdomain):** Map a CNAME in Cloudflare DNS pointing `dashboard.yourdomain.com` to `global-analytics-dashboard.pages.dev`.
- **Option B (Portfolio Subpath):** Add a Cloudflare Worker Route on your existing portfolio zone mapping `yourdomain.com/projects/global-dashboard/*` directly to the Pages target.

Project Milestone Timeline:

- **Week 1:** Data Pipelines & Edge Data Harmonization
- **Week 2:** Cloudflare Worker API & Workers AI Integration

- **Week 3:** shadcn/ui Component Development & Recharts Integration
- **Week 4:** Performance Audit, Subdomain Routing & Launch